

#	Milestone	Generative AI Guidelines	Due	Mark Release
3	3MT: Three-Minute Talk ¹	C: AI-Assisted Editing ²	Slide: Oct. 14, Wed., AoE³ Talk: Oct. 15, Thu., 8:30AM Vote: Oct. 15, Thu., AoE	Oct. 19, Mon.

This course is *project driven*. Students will work in teams to define, develop, and complete a project involving data modeling and representation over the course of one semester. The course project has two main objectives:

- Hands-on experience: Gain practical experience in analyzing and modeling real-world data, while developing skills in data representation, schema design, querying, implementation, evaluation, and teamwork.
- Research experience: Develop an understanding of the research *process*, from identifying a problem and reviewing existing approaches to designing, implementing, and evaluating different representations of data in the same domain.

Projects should aim to advance the state of the art, rather than simply reproduce an existing solution. This provides students with an opportunity to conduct meaningful project while completing the course requirements.

A data modeling project typically starts with a real-world domain and an information need. In this sense, a data modeling project may resemble a software or database project; however, there are important differences. A software project may primarily focus on implementing a functional system, while a data modeling project in this course focuses on understanding the structure and meaning of information and determining how that information can be represented, stored, and queried effectively for globally shared, accessible, and interoperable representations that can support Knowledge Graphs, the Giant Global Graph (GGG), and Linked Open Data⁴.

In this course, students will progressively explore different approaches to representing the same real-world domain. The project will begin with conceptual data modeling, including the identification of entities, attributes, and relationships and their representation using an Entity-Relationship Diagram (ERD). The conceptual model will then be transformed into a relational model (logical level of data modeling), including relational tables, keys, and constraints, and implemented and queried using SQL. The project will subsequently extend the representation toward semantic and knowledge representation, using RDF, RDFS, and OWL, and will explore the storage and querying of the resulting knowledge graph using a triple store and SPARQL:

Real-world domain → ERD → Relational Schema → SQL → RDF/OWL → Triple Store → SPARQL

The goal is not simply to implement these technologies independently. Rather, students should understand how different data models represent information and relationships, how one representation can be transformed into another, what information and semantics may be preserved or lost during such transformations, and how different representations affect the types of questions that can be answered.

¹ <https://www.uvic.ca/students/graduate/thesis-dissertation/three-minute-thesis/index.php>

² AI can be used to make improvements to the clarity or quality of student-created work to improve the final output, but no new content can be created using AI (e.g. language editing for lab reports). AI can be used, but your original work with no AI content must be provided in an appendix.

³ <https://www.timeanddate.com/time/zones/aoc>

⁴ <https://www.dbpedia.org/>

Roughly, the data modeling project in this course consists of the following milestones (phases):

1. Project Definition & Team Formation (Proposal)
2. Background & Related Work (Literature Review)
3. **Project Presentation: 3-Minute Talk (3MT)**
4. Solution Design & Implementation (Data Modeling & Representation)
5. Evaluation (Experimentation)
6. Project Demonstration: 10-Minute Talk (10MT)

A project manual has been prepared to guide students through each milestone. Each milestone breaks the project into a series of smaller steps and deliverables. The current manual covers Milestone 3: Project Presentation: 3-Minute Talk (3MT), also known as Elevator Pitch.

3.1. Presentation

The purpose of the 3MT is to communicate your project clearly and engagingly to an audience that may not be familiar with your domain or its technical terminology. You should tell a coherent story rather than present a condensed version of your report. Your presentation should address the following five questions:

1) *What is this all about? What is the story?*

Begin by introducing your domain and the problem your project addresses. Do not assume that the audience knows the technical terms, datasets, standards, or concepts used in your project. Imagine that you are explaining the project to a family member or a friend from a completely different field. Explain:

- What is the domain?
- What information are we dealing with?
- What is difficult or interesting about representing this information?
- What is the overall goal of your project?

Start with the real-world story, *not* with technical terminology.

2) *Why should we care?*

Now explain why this problem matters. Why is it useful to organize, represent, connect, or integrate this information? Who could benefit from a better representation of the domain? Your motivation could involve:

- Improving access to information;
- Connecting information from different sources;
- Reducing ambiguity or inconsistency;
- Enabling better search or querying;
- Supporting data integration;
- Improving interoperability between systems; or
- Enabling new applications through a knowledge graph.

Make the audience think: *"I understand why this information matters and why representing it well is important."*

3) *Who else tried to solve the problem in the past? How has this information been represented before?*

This part is a brief summary of background and existing data models. Explain how people, organizations, or systems have previously represented information in your domain. You may briefly discuss:

- Historical or traditional ways of recording the information;

- Existing databases or schemas;
- Standards or ontologies;
- Existing data models;
- Different representations used by different organizations or domains.

Focus on the main ideas and differences, rather than listing many papers, databases, or technical details. A useful question to answer is: *“How do existing approaches represent this domain, and how are their representations different?”*

Do *not* claim that your project is the first attempt to represent the domain simply because you have not yet found an existing model. Your literature and background review should help you understand how the domain has already been represented.

4) *What is missing from existing representations?*

In this part, you have to shine. This is where you explain the gaps, limitations, or opportunities that motivate your project. Based on your comparison of existing data models, identify one or two important issues that your project intends to address. For example:

- The important entity X or relationship Y is missing;
- Information from X, Y, Z sources cannot be easily connected;
- X, Y, Z models use inconsistent concepts or terminology;
- The important distinction between X and Y is not explicitly represented;
- The existing representation does not support the important query X;
- The model is too specific to X application and cannot generalize to Y closely related application.

Do not try to solve every problem you find. Select one or two meaningful gaps that are appropriate for the scope of your project. You should be able to say: *“Existing representations do X well, but they do not adequately represent Y. Our project therefore focuses on addressing Y.”*

5) *What is your proposed data model and knowledge-graph approach?*

In this part, you explain at a high level how your project addresses the identified gaps. Show your draft ER diagram or another simple visualization and explain the most important modeling decisions:

- What are the key entities?
- What are the important relationships?
- What information have you added or represented differently?
- How does your model address the gaps identified in the previous section?
- How might this model eventually be represented as a knowledge graph?

You do not need to explain every entity, attribute, or relationship in your ER diagram. Focus on the parts that are most important to your project's contribution. You should give the audience a clear roadmap from the problem to your proposed representation:

Domain → Existing Representations → Identified Gaps → Proposed Data Model → Knowledge Graph

The technical details of RDF, RDFS, OWL, SPARQL, implementation, or other technologies are *not* the focus of the 3MT unless they are essential to explaining your main idea.

Remember: Tell a Story. A successful 3MT is *not* a three-minute version of your report. It is a story that takes the audience through: What is the problem?→Why does it matter?→How is it represented today?→What is

missing?→What are you proposing? The audience should be able to understand the problem, motivation, gap, and main idea your proposed data model even if they have never taken a course in databases or Semantic Web.

3.2. Submission Guidelines

3.2.1. Deliverables

- Submission includes presentation slide and delivery of the talk in English. The presentation is limited to *exactly* 1 slide (also known as poster slide), no more and no less, including author list, figures, tables, and references, with unlimited pages for C: AI-Assisted Editing appendix. We understand that fitting all the required content into exactly one slide can be challenging. However, this is intentional: an important part of communication is learning how to retouch, refine, condense, and prioritize information without losing its essential meaning. We recommend initially creating a longer version (e.g., 3-5 slides) and then gradually reducing and retouching it until you can effectively communicate the key ideas within the required one-slide limit. You have to submit the slide a day ahead, i.e., **Wednesday Oct. 14, AoE**.
- The submission must be provided as a single zip file named: 3_3MT_firstname1_firstname2_[...].zip, e.g., 3_3MT_Kovvy_Hossein.zip, containing:
 - o The compiled pdf file for the one slide for presentation.
 - o Any required supplementary/appendices files.
 A *sample* is attached to this manual and is also available [here⁵ \(3_3MT.pdf\)](#).
- The presentation should be delivered in class before the other classmates on **Thursday Oct. 15, 8:30 AM**.

3.2.2. Team and Individual Contributions

- Each team must maintain a Git-based repository (e.g., GitLab, GitHub, ...) ⁶ for the project and linked inside the report. The repository must be associated with the individual accounts of all team members. The repository will be used to document the development of the project and to make individual contributions traceable. The repository should be used throughout the project rather than only immediately before a submission deadline. The instructor and/or marker may consider the repository's commit history, pull requests, issues, documentation, and other project records when assess individual contributions. The number of commits alone will not be used as a measure of contribution. The quality, significance, and nature of each member's contributions will also be considered.
- All team members are expected to make substantive contributions, and no member should be limited solely to coordination or project management.

3.2.3. Marking Scheme

- 10% What is this all about? What is the story?
- 10% Why should we care?
- 10% Who else tried to solve the problem in the past (old history if exist up to now)?
- 10% What is missing from existing representations?
- 10% What is your proposed data model and knowledge-graph approach
- 50% Average ranking by the audience and the instructor. An online poll will be provided and should be submitted by end of the day of presentation **Thursday Oct. 15, AoE**.

3.2.4. Sample Great Talks

- A Talk That All Parents Need to Hear: <https://fb.watch/iFrProhM8G/>
- McGill University, 3MT CHAMPION, 2022: <https://www.youtube.com/watch?v=hpdvfm2XWvk>

⁵ <https://www.overleaf.com/read/vtggnmssqqfqy>

⁶ Web-based applications and cloud services may collect, use and/or disclose your personal information and store or access that information outside of Canada. Personal information includes, but not limited to, identifiable individual's information such as your name or your email address. At UVic, GitLab at gitlab.uvic.ca is highly recommended for coursework, research and development projects. For more information, see <https://www.uvic.ca/systems/services/pages/gitlab.php>. At your team's discretion, you may use GitHub to store information about your project, but we do not require nor recommend storing any personal identifying information. Also, we do not require public repo, but it should be accessible to instructors and markers for evaluation purposes.